

Winds

- [Winds using Drift method](#)¹
 - Coordinate systems
 - Matrices and Matrix multiplication
- [Moment terms](#)²
- [Maneuvers for correction coefficients](#)³
 - [Other data](#)⁴

Links to files:

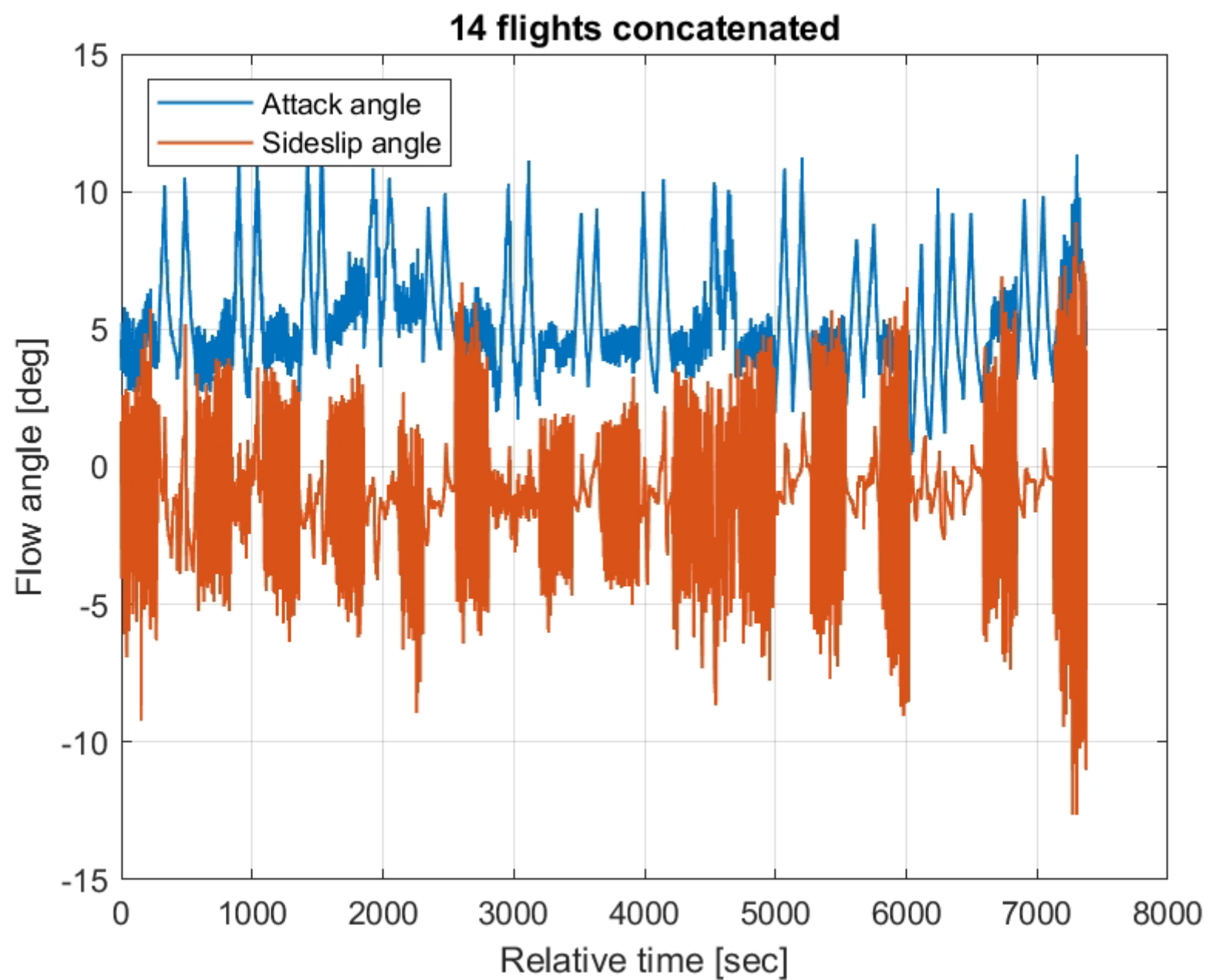
1. Wind.pdf
2. Moment-terms.pdf
3. WYO_maneuvers.pdf
4. Maneuvers.pdf

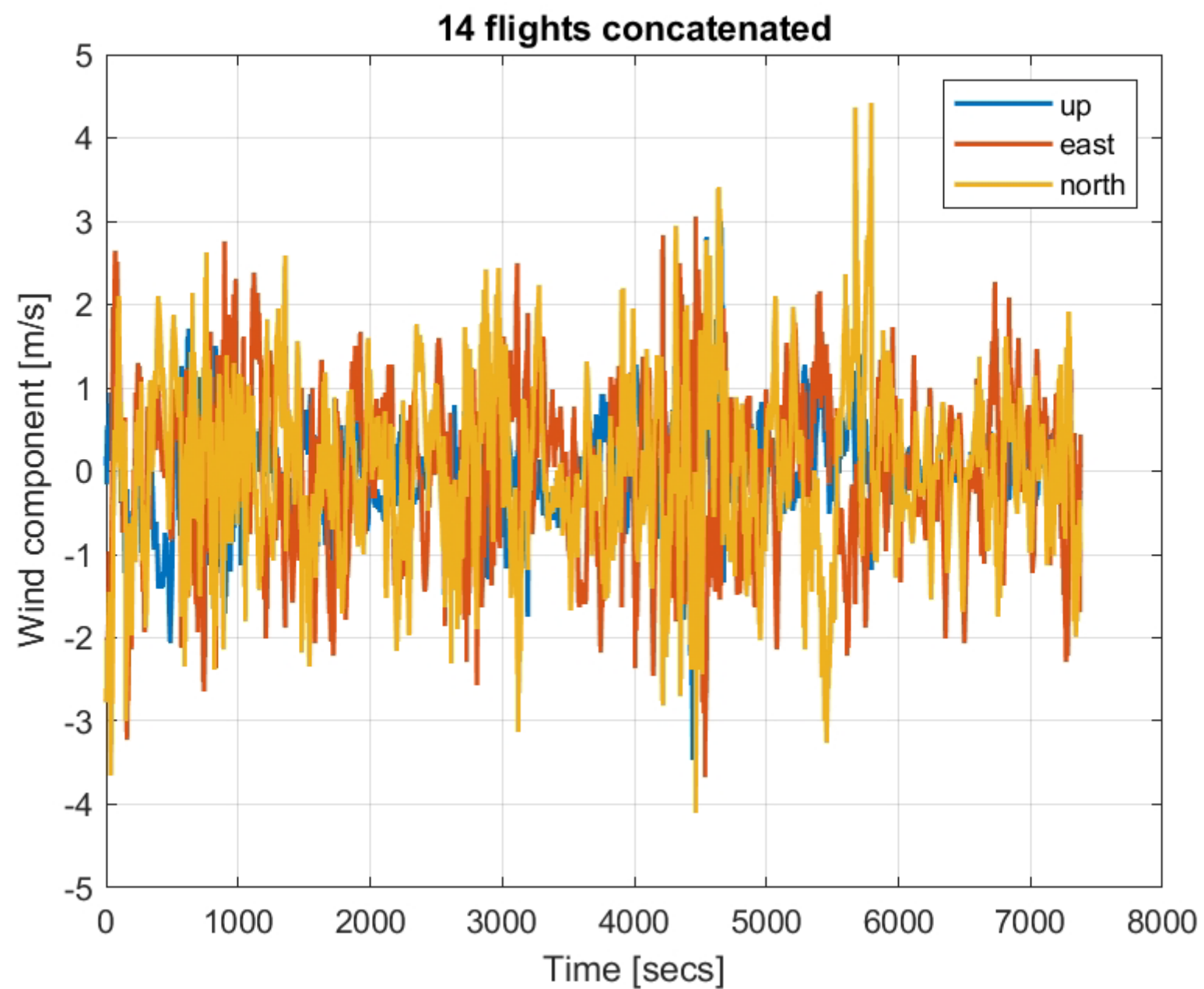
Maneuvers:

Flight in which pilot induces motions to vary inputs to wind calculation:

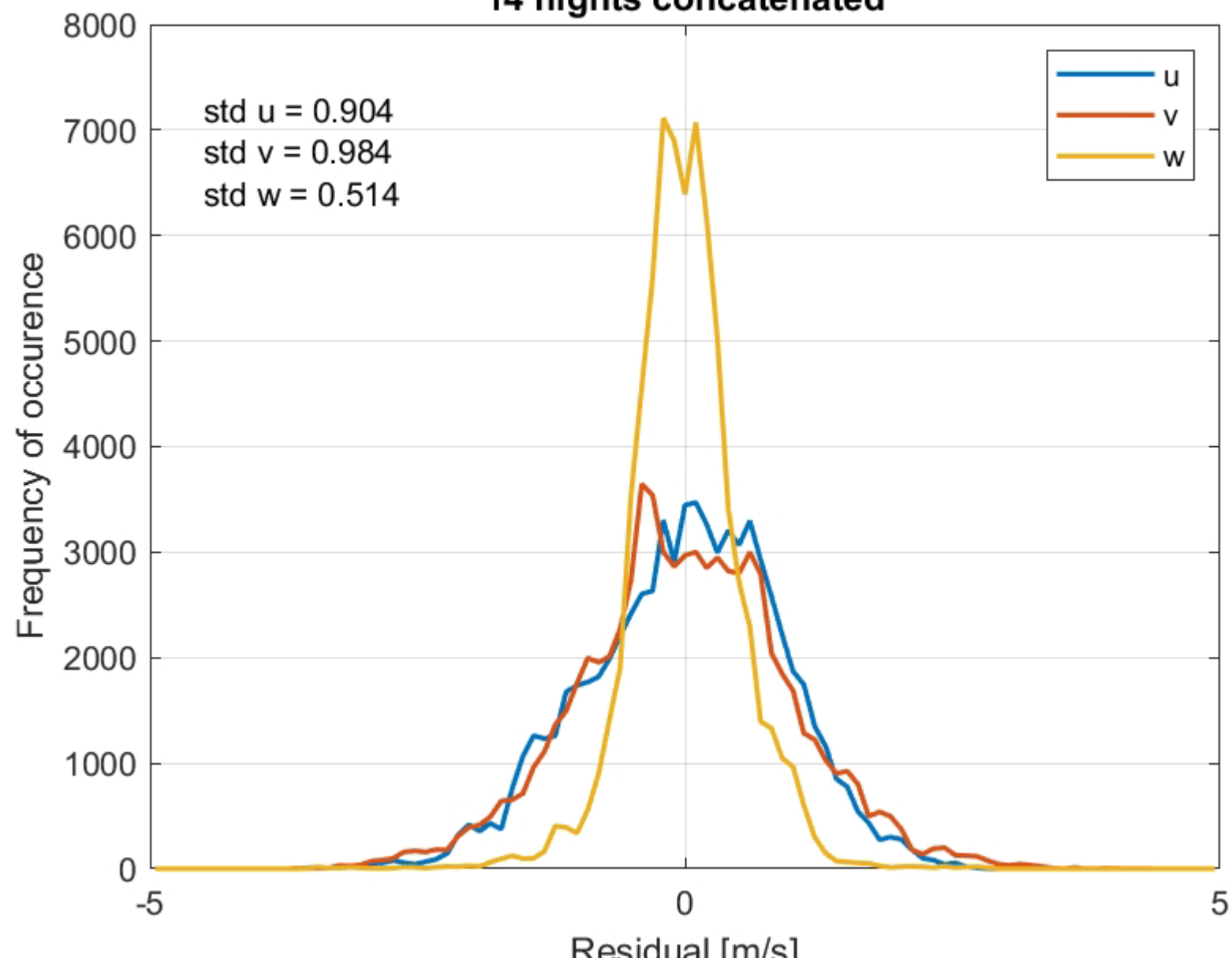
List of parameters:

1. Pitch offset (from flow angle boom)
2. Roll offset
3. Attack angle offset ($\alpha = a_off + a_frac * \alpha_m$)
4. Attack angle upwash fraction
5. Sideslip angle offset ($\beta = b_off + b_frac * \beta_m$)
6. Sideslip angle upwash fraction
7. Dynamic pressure offset
8. True heading offset

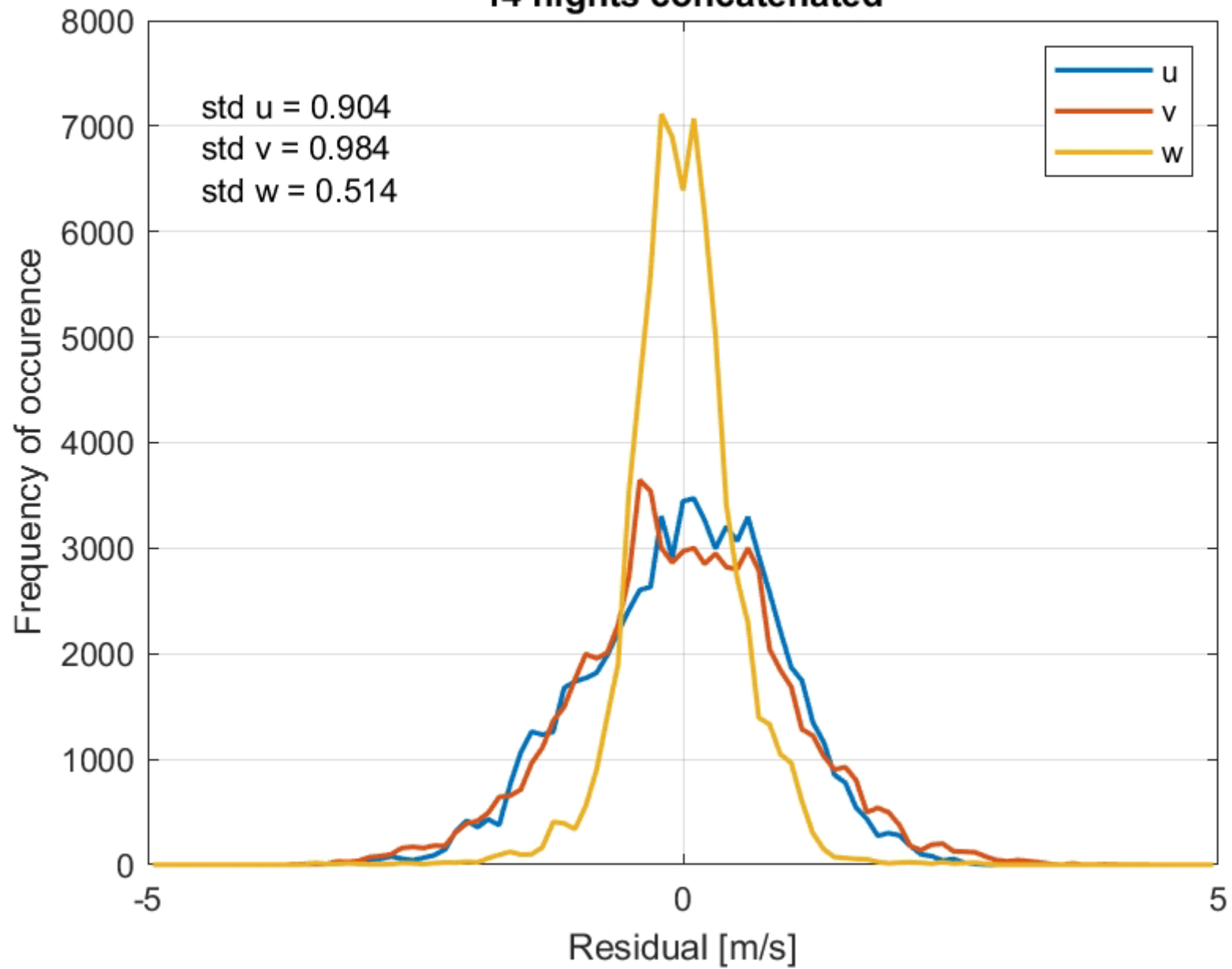




14 flights concatenated



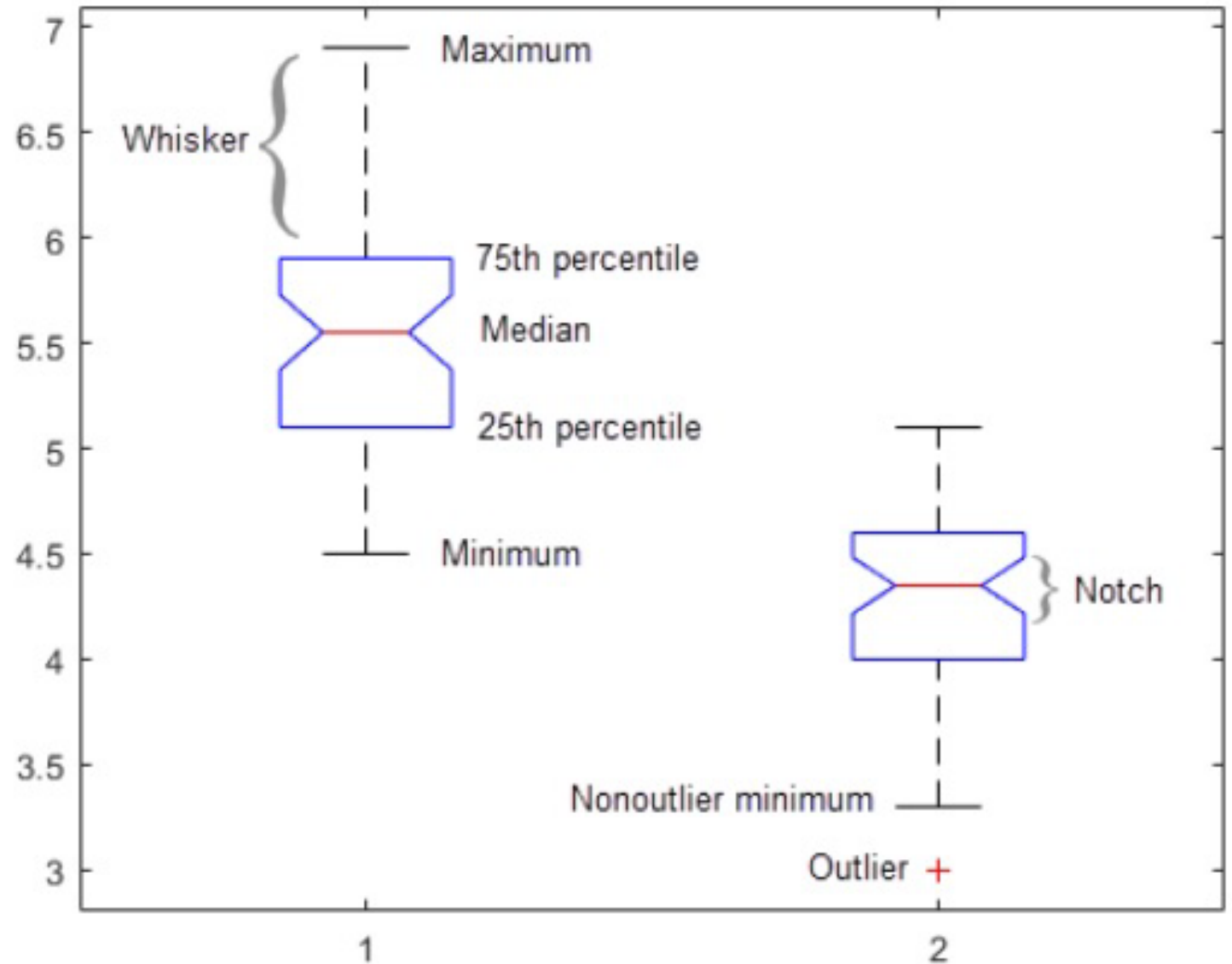
14 flights concatenated



Box and Whisker visualization

- Box is interquartile 25th – 75th percentile (50% of data)
- Whisker is 1.5 x interquartile, or $\pm 2.7 \sigma$, 99.3% of data)
- Outside Whisker are “outliers”

“Notch” shows most likely values expected for the median when the data represents a sample. If notches of 2 groups overlap, then they are from the same distribution with 95% confidence.



14 flights concatenated

